

SAJIB RAFI HOSSAIN

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Education

Queens College, City University of New York

Bachelor's in Computer Science

Expected May 2027

Queens, New York

Technical Skills

Languages: JavaScript · Python · TypeScript · Java · C++ · HTML · CSS

Technologies/Frameworks: React.js · Next.js · Tailwind CSS · Firebase · Node.js · Express.js · Drizzle ORM · PostgreSQL · SQLite · DynamoDB · AWS · Supabase · Vitest · Git/Github · Bash Command Line

Experience

Break Through Tech

March 2026 - Present

AI/ML Fellow

Remote

- Selected for a competitive AI/ML fellowship on machine learning, industry collaboration, and applied AI development
- Building hands-on experience with Python, pandas, NumPy, and scikit-learn through technical labs & coursework
- Preparing to pursue Agentic AI track in Spring '27 through advanced project-based training in applied AI systems

CUNY Tech Prep

July 2025 - Present

Software Engineer Fellow

Remote

- Selected for a competitive, year-long fellowship for top CUNY computer science students, focusing on full-stack web development, cloud technologies, and professional software engineering practices
- Practicing system design by translating requirements into scalable architectures, defining APIs, modeling data, and evaluating tradeoffs (performance, cost, reliability) through diagrams, written docs, and iterative design reviews
- Building full-stack applications with React/Next.js (TypeScript), Tailwind CSS, Node.js/Express, PostgreSQL, DynamoDB, Supabase, and AWS, supported by CI/CD pipelines with GitHub Actions
- Implementing engineering best practices, including minimum **70%** test coverage, Lambda API development, infrastructure testing with LocalStack, and robust documentation (ADRs, READMEs, API docs)

CodePath

September 2023 – April 2024

Web Developer Fellow

Remote

- Strengthened skills in HTML, CSS, JavaScript, and React through projects, including API integration and & parsing
- Collaborated in team-based coding sessions, improving project delivery through code reviews and pair programming

Projects

ClearSource | Next.js, TypeScript, Prisma, PostgreSQL, Clerk, RAG

April 2026

- Building ClearSource, a vegan product verification app that uses a retrieval-first RAG workflow to compare submitted product data against a verified knowledge base and return evidence-backed match explanations
- Creating a human-in-the-loop verification pipeline that flags uncertain ingredients, drafts manufacturer inquiries, supports admin-reviewed final decision, and future embedding/pgvector-based retrieval

GreenBites | TypeScript, React, Next.js, Supabase

December 2025

- Developed a full-stack nutrition and sustainability tracker that lets users search foods via **Calorie Ninjas API**, adjust serving sizes, and track daily macros, calories, and carbon footprint with interactive dashboards
- Designed **PostgreSQL** database schema on Supabase with **Drizzle ORM**, protected routes with Clerk authentication, resolved production deployment issues, and fixed persistence issues so tracking remains accurate across sessions

CUNY GradeLens | JavaScript, React, Node.js, Express.js, OpenAI API, SQLite, Tailwind CSS

August 2025

- Built an AI-powered platform during the CTP Hackathon as part of a team of 3 that analyzes grade distributions with GPT-4, enabling **natural language queries** to deliver insights on professors and course difficulty
- Developed a chatbot with conversation history, fuzzy name matching algorithms, and data visualizations (**Chart.js**), robust **CSV data ingestion**, regex-based entity extraction, and multi-level caching to optimize API costs by **60%**

Leadership / Community Impact

STEM Instructor

July 2025 - August 2025

New York Edge

- Led environmental science and STEM lessons (sustainability, renewable energy, ecosystems, etc.) to **150+** grade 3–5 students; adapted instruction to diverse learners and fostered engagement through interactive, hands-on activities.